Chapter 3 exercises

DS03-01 • Circular wraparound

Use head and size with capacity C=4. From the empty state `(head,size)=(0,0)`, enqueue A,B,C, dequeue twice, enqueue D,E,F, then dequeue three times. Give head, size, logical sequence, and physical positions after every operation.

DS03-02 • Expression conversion

Convert `(9-4)*(2+3)` to prefix and postfix. Draw its expression tree and evaluate it with a stack. Explain why the pop order for postfix `8 3 -` cannot be exchanged arbitrarily.

DS03-03 • The first DFS path

Design an unweighted graph of at most six vertices and a neighbor order such that DFS first reaches the target along a longer path than BFS finds. List both visit orders and predecessor assignments.

Hint. Give the start both a short path and a longer path that DFS explores first.

DS03-04 • Syntax and safe evaluation

At what conditions must postfix inputs `2 +`, `2 3`, and `8 0 /` be rejected? For signed 64-bit values, what arithmetic checks are needed besides division by zero? A complete lexer is not required.

DS03-05 • End-of-input and grammar

Convert `9-4*2/1+3`, listing every token, operator stack, and output, including EOF. Give rejection points for `(9-4`, `9-4)`, and `9-`. Compare postfix forms and values for left- and right-associative `2^3^2`.

DS03-06 • Queue growth and the abstract value

Capacity is 4 with physical=[E,F,C,D],head=2,size=4. Grow to 8. List old and new slots for every logical value and the committed head/size. Refute direct slot copying with retained head using an actual accessed position. Analyze one growth and repeated doubling.

References and attribution

Ellis Horowitz, Sartaj Sahni, Dinesh Mehta. Fundamentals of Data Structures in C++, 2nd Edition.

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https://github.com/xinhi6716/computer-science-self-study